


# Pulpal Considerations

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Because caries is a bacterial infection, it has a deleterious effect on the pulp, ranging from mild inflammation to pulpal death. In addition, virtually all restorative procedures cause pulpal irritation. As discussed in chapter 1, the pulp has inherent defense mechanisms to limit damage caused by irritants. There are also a number of dental procedures performed with the goal of preserving pulpal health. Most of these procedures attempt to provide a barrier to external irritants by placing a protective sealer or liner on the cavity walls.

Before placing or cementing a restoration into a cavity preparation, the clinician must decide if a protective cavity base, sealer, or liner should be placed. While seemingly simple, this decision has been complicated by an ever-increasing number of products for sealing and lining. This chapter reviews pulpal considerations relevant to operative dentistry, including the effects of cavity preparation, caries, and restorative materials on the pulp. In addition, it defines the various protective materials, describes the ways they interact with and provide protection for the pulp, reviews the properties of current materials, and discusses the changes that have occurred in this area of operative dentistry in recent years.

# Physiologic Considerations

The physiology of the pulp is influenced by several factors that form the basis for the decision to use a sealer, liner, and/or base.

## Remaining dentinal thickness

No material that can be placed in a tooth provides better protection for the pulp than dentin. Dentin has excellent buffering capability to neutralize the effects of cariogenic acids,[1] and it insulates the pulp from temperature increases during cavity preparation.[2] The remaining dentinal thickness (RDT), from the depth of the cavity preparation to the pulp, is the single most important factor in protecting the pulp from insult.[1,3] In vitro studies have shown that a 0.5-mm thickness of dentin reduces the effect of toxic substances on the pulp by 75% and a 1.0-mm thickness reduces the effect of toxins by 90%.[4] Little pulpal reaction occurs when there is an RDT of 2 mm or more.[1] The greatest impact on the pulp occurs when the RDT is no more than 0.25 to 0.30 mm.[3,5,6] Conservation of remaining tooth structure is more important to pulpal health than is replacement of lost tooth structure with a cavity liner or base.

## Causes of pulpal inflammation

Like other soft tissues, the pulp reacts to an irritant with an inflammatory response.[7,8] It was previously believed that pulpal inflammation was primarily the result of the toxic effects of dental materials.[9,10] More recent evidence, however, demonstrates that pulpal inflammatory reactions to dental materials are mild and transitory; significant adverse pulpal responses occur more as the result of pulpal invasion by bacteria or their toxins[3,11–18] (Fig 6-1a). Even early enamel caries lesions that extend less than one-fourth of the way to the dentinoenamel junction (DEJ) have been shown to induce a slight pulpal reaction, particularly when the caries lesion has advanced rapidly. This is probably due to an increase in the permeability of enamel, allowing the transmission of stimuli along enamel rods.[19,20]

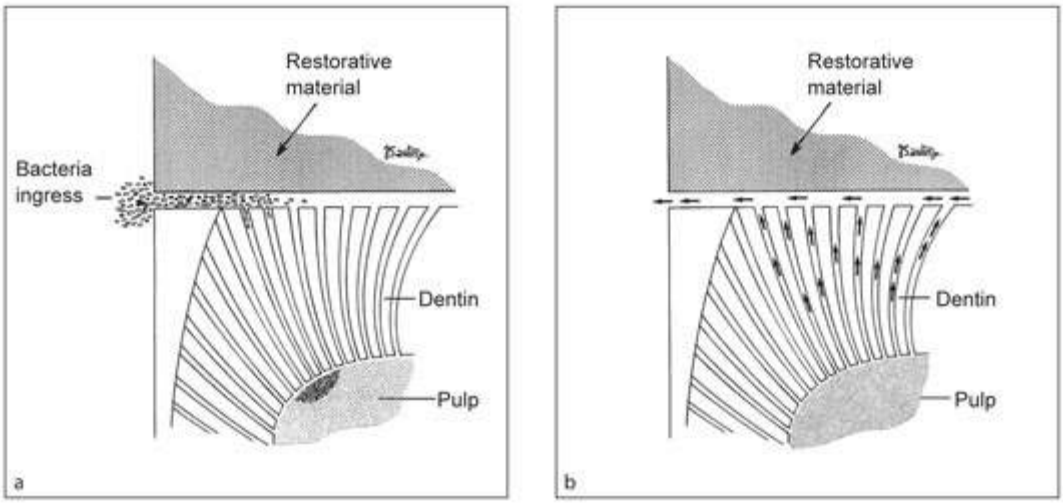



**Fig 6-1** *(a)* Bacteria will penetrate the marginal gap and dentinal tubules from the saliva, which may cause pulpal irritation, pulpal necrosis, or recurrent caries. *(b)* If a restoration is not well sealed, fluid flows out of the dentinal tubules and into the space between the restorative material and the tooth surface *(arrows)*. A stimulus such as heat or cold causes a change in the flow rate, which is interpreted by mechanoreceptors as pain.

As a lesion progresses deeper into the tooth, pulpal reaction increases.[19] When actual pulpal encroachment by bacteria and/or their toxins has taken place, severe inflammation or pulpal necrosis frequently occurs.[14] The outward flow of fluid through dentinal tubules does not prevent bacteria or their toxins from reaching the pulp and initiating pulpal inflammation.[21] The caries process also induces the formation of reparative dentin and reactive dentin sclerosis, which increases the protective effects of the remaining dentin.[22]

When bacterial contamination is prevented, favorable responses have been found in pulpal tissue adjacent to many restorative materials. Those materials include amalgam, light-activated resin composite, self-curing resin composite,[23] zinc phosphate cement, silicate cement,[13] glass-ionomer cement, and acrylic resin.[24] Acid etching of dentin has long been considered detrimental to the pulp, but the pulp can readily tolerate the effects of low pH if bacterial invasion is prevented[12,23,25] and resin components are precluded from traversing the dentinal tubules and entering the pulp.[5,26]

A number of instrumentation techniques elicit pulpal responses as well. The most common are rotary instruments used in high- and low-speed handpieces for tooth preparation. Tooth preparation can be traumatic to the pulp, and a number of factors affect pulpal reaction. The degree of pulpal reaction is dependent on the amount of friction and desiccation.[27,28] One retrospective study indicated that high-speed preparation, with the use of light force (1 to 3 oz), new burs, air coolant, and

intermittent water spray from the air-water syringe, rarely led to a need for postrestoration endodontic therapy.[29] However, the key to controlling both friction and desiccation is water spray at the site of contact between the bur and tooth structure. While some research has shown that this is more important than the amount of water that is used on a rotating bur,[28] another study has demonstrated that increased water spray volume can significantly reduce pulpal temperature rises caused by friction.[2] Frictional heat generated by tooth preparation can result in burn lesions in the pulp and abscess formation.[30] While it is often advantageous to refine aspects of a cavity preparation without water spray in order to aid visibility, this must be done conservatively. The pulp can tolerate dry preparation in a limited area, but the severity of the pulpal reaction increases as the area of dentin subjected to preparation without water spray increases.[28] Another consequence of desiccating dentin in a preparation is that dentinal fluid is lost from the tubules. The lost fluid may be replaced with chemicals that can elicit a harmful pulpal reaction.[28]

The temperature rise is considerably greater when enamel or a combination of enamel and dentin is prepared versus preparation of dentin alone.[31] Additionally, research has shown that pressure applied during rotary instrumentation has a greater effect on temperature rise than does rotational speed,[31] which is probably why preparation using low-speed rotary instrumentation has been shown to be more traumatic to the pulp than high-speed preparation.[30] Diamond burs tend to produce more temperature increase than do carbide burs,[30] and the reaction of the pulp tends to increase as the depth of the cavity preparation increases (and RDT decreases).[32,33] Considering the latter two findings, it should not be surprising that an occasional consequence of full-coverage restorations is pulpal necrosis.[34–36] Research has shown that 3% to 22% of teeth with full-coverage crowns require endodontic therapy.[35–37] It is clear that the combination of reducing water coolant and RDT and increasing bur speed and pressure can produce a significant increase in temperature during tooth preparation.[2,30,31,38] Therefore, the keys to minimizing adverse pulpal reaction from rotary instrumentation are the following:

- Adequate air-water coolant spray
- Light pressure
- Sharp rotary cutting instruments
- Preservation of tooth structure

Two relatively new methods for tooth preparation are available—lasers and

kinetic cavity preparation, also known as *air abrasion*. Animal studies have shown that air-abrasion cavity preparation is no more traumatic to the pulp than rotary instrumentation.[38,39] Likewise, the use of a variety of lasers, including carbon dioxide, erbium-doped yttrium aluminum garnet, neodymium-doped yttrium aluminum garnet, erbium-chromium–doped yttrium scandium gallium garnet and free-electron lasers, on tooth structure has demonstrated minimal pulpal response compared with that of high-speed rotary instrumentation.[40–44] The key to minimizing thermal pulpal insult during tooth preparation with a laser, as with preparation using rotary instrumentation, is the use of water coolant.[45]

One other modality commonly used in conjunction with operative dentistry—electrosurgery to remove gingival tissue for enhanced access during tooth preparation and impression making—may affect the dental pulp. Several animal studies have shown that as long as the contact of the electrosurgical probe is with intact enamel, little or no pulpal reaction ensues. However, if the probe contacts a metallic restoration, adverse and often severe pulpal reaction results.[46–49] This adverse reaction occurs regardless of whether a cavity base is present.[49] The pulpal response is more severe with increased contact time (longer than 0.4 seconds)[47] and decreased dentin thickness between the metallic restoration and the pulp.[46,49]

## Causes of pulpal pain

The causes of pulpal pain and sensitivity, while not fully explained, are becoming better understood. Increased intrapulpal pressure on nerve endings, secondary to an inflammatory response, is one mechanism that may explain pain as a result of bacterial invasion.[1] However, this interpretation fails to explain sensitivity that occurs in the absence of inflammation. The explanation for pulpal pain in the absence of inflammation that is most accepted is the hydrodynamic theory, which is described in detail below.[50]

## Causes of thermal sensitivity

Prevention of postoperative thermal sensitivity has long been a rationale for the placement of cavity bases beneath metallic restorations. Initial in vivo research documenting the alleged problem was sparse and poorly controlled. Although one study showed reduced postoperative sensitivity when thick cement bases were used,[51] another demonstrated that, by 6 months postoperatively, few patients had thermal sensitivity regardless of whether a cavity base had been placed.[52] In one

survey, 50% of patients questioned 24 hours after restoration placement reported some discomfort, but 78% of these patients described the discomfort as mild and fleeting.[53] Several more recent studies have demonstrated that a significant majority of patients receiving amalgam restorations do not experience postoperative thermal sensitivity, regardless of lesion depth or the absence or presence of a particular cavity sealer or liner. Of those patients with postoperative thermal discomfort, almost all describe it as minor, and it has almost always disappeared within 30 days.[54–60] Any discussion of the need for protection against postoperative thermal sensitivity must be tempered by the understanding that the prevalence and magnitude of this problem have likely been overestimated.

## *Theory of thermal shock*

There are two theories about the cause of thermal sensitivity (usually to cold) following restoration placement and, consequently, two philosophies about how to best address the problem. The first theory states that sensitivity is the result of direct thermal shock to the pulp via temperature changes transferred from the oral cavity through the restorative material,[61,62] especially when remaining dentin is thin. Protection from this insult would then be provided by an adequate thickness of an insulating material with low thermal diffusivity.[62,63] It has been noted that resin composite exhibits such low thermal diffusivity that a thermal insulating base should be unnecessary in conjunction with resin composite restorations.[63,64] Use of an insulating base for thermal protection would therefore be limited to metallic restorative materials that exhibit higher rates of temperature transfer.

When a base is used to provide insulation to counter thermal sensitivity in amalgam restorations, the thickness of the material must be minimized in areas subject to occlusal loading. Research has shown that, as the thickness of the base increases, the fracture resistance of the overlying amalgam decreases.[65,66] Because temperature diffusion through amalgam to the floor of the cavity preparation is effectively reduced by 0.50 to 0.75 mm of basing material, if a base is used, its thickness should be restricted to no more than 0.75 mm.[63] Modulus of elasticity is the key property that determines how effectively a base or liner will support an amalgam restoration; a high modulus of elasticity indicates stiffness, while a low modulus of elasticity indicates flexibility. As the modulus of elasticity of a basing material decreases, the resistance to fracture of overlying amalgam decreases.[65–67]

## *Theory of pulpal hydrodynamics*

The more widely accepted theory of thermal sensitivity holds that temperature sensitivity is based on pulpal hydrodynamics. Most restorations have a gap between the wall of the preparation and the restorative material that allows the slow outward movement of dentinal fluid (Fig 6-1b). Cold temperatures cause a sudden contraction of this fluid, resulting in a rapid increase in the flow, which is perceived by the patient as pain.[50] As dentin nears the pulp, tubule density and diameter increase,[1,68] as does permeability,[69] thus increasing both the volume and the flow of pulpal fluid susceptible to the hydrodynamic effects of cold temperatures. This may explain why deeper restorations are sometimes associated with more problems of sensitivity.[17]

According to this theory, if the tubules can be occluded, fluid flow is prevented and a cold temperature does not induce pain. The operative factor in reducing sensitivity to thermal change thus becomes effective sealing of dentinal tubules rather than placement of an insulating material of a certain thickness.[50] Scanning electron microscopic observations have revealed significantly higher numbers of open tubule orifices in hypersensitive dentin, lending credence to this theory.[70,71]

The theory of pulpal hydrodynamics has gained general acceptance in recent years and has changed the direction of restorative procedures away from thermal insulation and toward dentinal sealing. Thus, there is increasing emphasis on the integrity of the interface between restorative material and prepared tooth.

## Cavity Sealers, Liners, and Bases

The terms *varnish*, *sealer*, *liner*, and *base*, used to describe a variety of materials, have been a source of confusion in dental literature. In 1995, McCoy[72] provided the following definitions for these terms.

- *Cavity sealers:* Materials in this category provide a protective coating to the walls of the prepared cavity and a barrier to leakage at the interface of the restorative material and the walls. The term *sealer* implies total prevention of leakage, but, in fact, the barrier provides various degrees of seal. Sealers usually coat all walls of a cavity preparation. Commonly used sealers fall into two categories:
  - *Varnish:* A natural rosin or gum (such as copal), or a synthetic resin, dissolved in an organic solvent such as acetone, chloroform, or ether.[73]
  - *Adhesive sealers:* Adhesive systems designed to provide sealing as well as bonding at the interface between restoration and cavity-preparation walls.

- *Cavity liners:* Cement or resin coating of minimal thickness (usually less than 0.5 mm) to achieve a physical barrier to bacteria and their products and/or to provide a therapeutic effect, such as an antibacterial or pulpal anodyne effect. Liners are usually applied only to dentin cavity walls that are near the pulp.
- *Cavity bases:* Materials to replace missing dentin, used for bulk buildup and/or for blocking out undercuts in preparations for indirect restorations.

Knowledge of the properties and indications for the materials in each category will aid in the selection process when the practitioner is faced with an array of choices.

## Cavity sealers

Cavity sealers provide a protective coating for freshly cut tooth structure of the prepared cavity. The tooth-restoration interface has always been considered critical in dentistry, a fact apparent in the profession's emphasis on marginal adaptation of dental restorations. The concern is that any interfacial gap, even one not readily apparent under magnification, will allow microleakage. Kidd[74] defined *microleakage* as the passage of bacteria, fluids, molecules, or ions along the interface of a dental restoration and the wall of the cavity preparation. This process is theorized to cause marginal discoloration, secondary caries, and pulpal pathosis. A summary of some clinical studies on the percentage of clinical failures of existing restorations due to secondary caries is provided in Table 6-1.

**Table 6-1** **Tooth-restoration interface: Materials and clinical failures***

| Study | Composite | Amalgam | Glass ionomer |
|---|---|---|---|
| Opdam et al[75] | 43% | 23.5% | NA |
| Bernardo et al[76] | 88% | 66% | NA |
| Soncini et al[77] | 51.8% | 43.6% | NA |
| Opdam et al[78] | 38% | 29% | NA |
| Forss and Widström[79] | 38.6% | 50% | 42.4% |

*Numbers denote percentage of failures attributable to secondary caries. NA–not applicable.

Clearly, the junction between the restorative material and tooth structure is the source of a considerable number of restoration failures; providing a seamless transition from restoration to tooth structure has long been a goal in dentistry. Cavity sealers, used to fulfill this function, take two forms: varnishes and adhesive sealers.

## *Varnishes*

As previously discussed, varnish is a natural gum (such as copal), a rosin, or a synthetic resin dissolved in an organic solvent, such as acetone, chloroform, or ether, that evaporates, leaving behind a protective film.[73] It is used as a barrier against the passage of bacteria and their by-products into dentinal tubules, and it reduces the penetration of oral fluid at the restoration-tooth interface. This film is very thin, usually 2 to 5 μm, and provides no thermal insulation.[80–82]

Copal varnishes were used for many years to fill the gap at the amalgam-tooth interface until corrosion products formed to reduce the gap.[80,83,84] Varnishes have also been used as barriers against the passage of irritants from cements and bacteria into dentinal tubules.[80] Two applications have been shown to be more effective than a single coat, but a third application does not significantly improve the coating of the cavity walls.[85] Copal varnish is capable of reducing dentin permeability by 69%[86] and significantly reducing microleakage for 4 to 6 months.[87–89] Although its use has diminished in recent years, varnish has commonly been used under amalgam restorations and before cementation of indirect restorations with zinc phosphate cement. Placement of copal varnish before crown cementation with zinc phosphate does not have a detrimental effect on retention.[90]

## *Adhesive sealers*

The most recent materials to be used as cavity sealers have a demonstrated multisubstrate bonding ability that allows the restorative material to adhere to tooth structure. Examples include adhesive bonding systems, resin luting cements, and glass-ionomer luting cements. The benefits of using adhesive bonding systems (described in chapter 9) to attach resin composite materials to tooth structure are well documented and accepted. It is well established that acid etching will promote a reliable, durable bond to enamel.[12] Its mechanism of action (ie, the diffusion of polymerizable monomers into porosities and channels established in enamel and dentin as a result of the demineralizing action of acid) is well accepted. Bonding systems also provide a chemical bond between the unfilled resin of the adhesive system and the resin composite. Enamel's more consistent and highly mineralized

structure provides a more reliable bond than that achieved to dentin.[91,92]

Researchers and clinicians have worked to develop cavity sealers that can improve the seal provided by cavity varnishes for amalgam restorations. Some studies have demonstrated that varnishes reduce, but do not eliminate, microleakage around amalgam restorations,[87,93,94] while other studies have shown no benefit or even increased microleakage when a varnish is used.[95–97] Because of postplacement amalgam marginal leakage, the duration of which is prolonged when the slower-corroding high-copper alloys are used, more effective and longer-lasting sealers have been sought. This has led to the frequent use of adhesive resins in conjunction with amalgam restorations.[98] At least one study has shown an adhesive to be inferior to varnish in the seal it provides99; others have shown adhesives and varnishes to exhibit similar degrees of microleakage, 100,101 while still others have shown adhesive resin to impart no greater seal than when no sealer at all is used.[95,102] However, numerous studies have shown resin adhesives to provide a significant reduction in leakage.[88,89,103–111]

While not unanimous, there is compelling evidence that adhesive resins used as sealers reduce interfacial microleakage compared with either unsealed or varnish-sealed amalgam restorations when evaluated in the short term (24 hours to 14 days). Superior sealing of dentinal tubules by bonding resins compared with varnish has also been demonstrated.[89,103,108,111] Animal research has demonstrated that dentin primers alone or in conjunction with dentin adhesives can significantly reduce dentin sensitivity[112] and that they have good pulpal compatibility when used in cavity preparations.[113,114] However, modern adhesives continue to exhibit leakage when cavity margins are on dentin or cementum.[92,115–118] Most in vitro research on the use of bonding systems with amalgam restorations is of short duration, and uncertainty exists concerning the durability of the bonded interface between amalgam and tooth structure.

Studies have found that both interfacial and dentinal tubule leakage increase significantly over periods of 1 month to 1 year after placement of the resin bonding/sealing resin and amalgam.[90,104,108,109]

Most important, numerous controlled clinical trials have failed to demonstrate a decrease in postoperative sensitivity with the use of adhesive agents under amalgam restorations compared with the use of either traditional sealers and liners or no cavity sealer at all.[54,56–59,119–122] These results are consistently found regardless of cavity depth and RDT.[57,58] Given these facts, there are some concerns about the use of adhesive resins under amalgam restorations. The insoluble adhesive layer may act

as a barrier to prevent amalgam corrosion products from ultimately sealing the tooth-restoration interface. As a result, the dentin bonding resins may potentially put the patient at greater risk for marginal leakage and recurrent caries in the long term. In addition, bonding resins are much more technique sensitive than varnishes,[118,123] and bonding systems are more expensive and time-consuming.[124] Researchers have also noted the tendency of self-curing adhesive sealers used in conjunction with amalgam restorations to spread to adjacent tooth surfaces, potentially giving rise to periodontal irritation.[59] Additional potential drawbacks to the use of adhesive sealers with amalgam include pooling of resin, resulting in radiographic artifacts,[125] and incorporation of sealer into the amalgam during condensation,[126] leading to significant loss of amalgam strength.[127,128] These considerations need to be weighed against clinical research evaluating the performance of bonded amalgam restorations. A number of studies have demonstrated clinical performance of bonded restorations that is comparable with that of nonbonded restorations.[54,59,120,122] One trial demonstrated equal performance of complex amalgam restorations (including at least one proximal surface and one cusp), using either pin retention or adhesive bonding over 6 years.[122] Those outcomes are encouraging because bonded amalgam restorations permit minimal tooth preparation rather than necessitating the removal of sound tooth structure to develop traditional resistance and retention form. However, the results of these clinical studies should be interpreted with some caution, because most have evaluated relatively low numbers of restorations and, with the exception of the 6-year study of large restorations cited above, have been of short duration (3 years or less) and have tested mostly small to moderate-sized restorations. With these precautions in mind, it should be noted that evidence is accumulating to support the use of adhesive sealers in conjunction with amalgam restorations.

## Cavity liners

Cavity liners are placed with minimal thickness, usually less than 0.5 mm, and provide some type of therapeutic benefit, such as fluoride release, dentinal seal through adhesion to tooth structure, and/or antibacterial action that promotes pulpal health.[72]

### *Calcium hydroxide*

Calcium hydroxide [$Ca(OH)_2$] has long been used as a liner because of its pulpal

compatibility and purported ability to stimulate reparative dentin formation with direct pulpal contact.[1] However, research has shown that not all formulations of calcium hydroxide have a stimulatory effect on human pulpoblasts.[129] There is a growing belief that reparative dentin formation is assisted, rather than stimulated, and that this is due to the antibacterial action of calcium hydroxide, which reduces or eliminates the inflammatory effects of bacteria and their byproducts on the pulp.[14,50,130,131] Research has also indicated that calcium hydroxide may release growth factors from dentin that can assist in pulpal healing.[131,132]

Conventional formulations of calcium hydroxide liners have demonstrated poor physical properties.[65,133] High solubility of some calcium hydroxide liners may lead to contamination of bonding resins and result in increased marginal leakage.[134] High solubility may also result in softening of the liner and in material loss under poorly sealed restorations.[50,135] Visible light–activated calcium hydroxide products have overcome most of these deficiencies. They exhibit improved physical properties[136] and significantly reduced solubility.[133] While the modulus of elasticity of the light-activated products has been shown to be higher than that of conventional calcium hydroxide in one study,[137] in another it was lower,[136] with a resulting reduced ability to support an overlying amalgam restoration.[138] These unfavorable physical properties restrict calcium hydroxide use to application over the smallest area that would suffice to aid in the formation of reparative dentin when a known or suspected pulp exposure exists.

### *Glass ionomer*

Glass ionomer has been used as a cavity liner in an attempt to take advantage of two highly desirable properties: chemical bond to tooth structure and fluoride release.[133] Although fluoride release from glass ionomer decreases with time,[139] sustained release has been demonstrated[140] with corresponding uptake into adjacent tooth structure.[141] This is thought to aid in anticariogenic activity.[142] Like zinc phosphate, glass ionomer is quite acidic on initial mixing but tends to neutralize within 24 hours.[143] Pulpal response to both visible light–activated and conventional glass-ionomer formulations has been shown to be favorable when not in direct pulpal contact,[23,144–146] likely because glass ionomer decreases interfacial bacterial penetration.[6,17,18,145] The exact mechanism by which this is achieved is uncertain, but it may be due to one or more of the following: fluoride release, initial low pH,[147] chemical bond to tooth structure (physically excluding bacteria),[145] or release of a

metal cation.[148,149] Both visible light–activated and conventional glass-ionomer liners exhibit good physical properties, with the conventional version exhibiting reduced interfacial gap formation,[150] a higher modulus of elasticity,[151] and subsequently improved support for amalgam restorations.[138] Glass ionomer has been shown to reduce microleakage under amalgam restorations.[95,152] Conventional glass ionomers are relatively soluble in an acidic environment and are susceptible to rapid surface deterioration when subjected to acid etching.[153] Visible light–activated glass ionomers show improved resistance to acid solubility[136] while maintaining fluoride release and bond to tooth structure.[151] Therefore, the visible light–activated formulations are more desirable for use with resin composite restorations.

Glass-ionomer cements have been recommended as liners under resin composite restorations to reduce microleakage. The use of glass-ionomer cement as an intermediate layer between dentin and resin composite, particularly in Class 5 restorations, is often referred to as the *sandwich technique*. Glass ionomer use, most often in conjunction with Class 2 resin composite restorations, is sometimes called the *bonded-base technique*. Both techniques can be "open," in which the glass-ionomer cement at the gingival margin is exposed, or "closed," in which the glass-ionomer cement is completely covered by resin composite. Glass-ionomer cement liners, both visible light–cured and self-cured, have been studied extensively for their ability to seal the interface between resin composite and the cavity preparation. The preponderance of in vitro evidence indicates that glass-ionomer cement liners perform at least as well as, and in most cases significantly better than, bonding resins used alone to seal the restoration-tooth interface. This is probably due to the chemical bond to tooth structure, as well as a delayed set and increased strain capacity provided by the glass-ionomer cement. This may reduce the tendency for polymerization shrinkage stress to pull the resin composite away from the preparation walls.[154] In addition, the open sandwich or bonded-base restoration appears to be superior to the closed technique in achieving this seal.[155–164] The use of the open sandwich technique with posterior resin composite restorations is discussed in greater detail in chapter 11.

## Cavity bases

As previously stated, cavity bases are used as dentin replacement materials, allowing for less bulk of restorative material or blocking out undercuts for indirect restorations[72] (Fig 6-2). Although cavity bases generally are not used for pulpal protection or health, they are briefly described here.

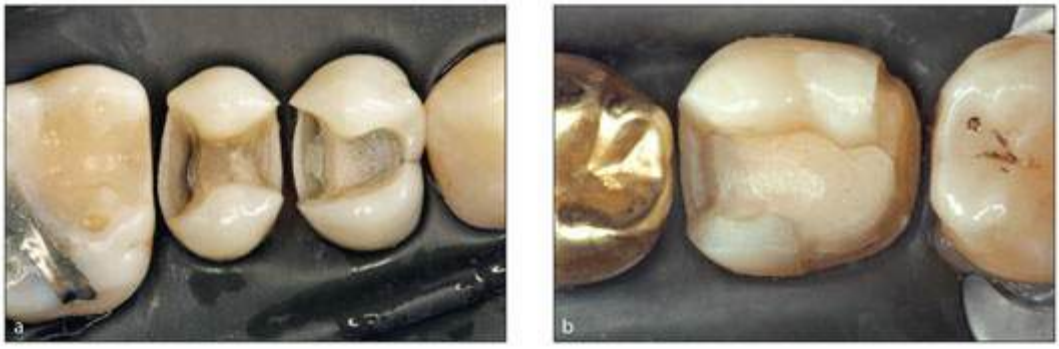


**Fig 6-2** The primary use for bases is to block out undercuts in divergent preparations. Glass-ionomer bases are used here to block out undercuts in ceramic inlay preparations *(a)* and in a gold onlay preparation *(b)*.

### *Zinc oxide–eugenol and zinc phosphate cements*

Zinc oxide–eugenol (ZOE) and zinc phosphate cements have been used for a number of years as bases under a variety of restorative materials. Although both provide excellent thermal insulation[13,63] and zinc phosphate cement exhibits excellent physical properties,[1] their use has diminished in recent years with the growing question of their benefit to pulpal health and with the advent of materials that release fluoride and adhere to dentin.[165]

### *Glass ionomer*

As previously discussed, glass-ionomer materials have excellent mechanical properties, modulus of elasticity, and restoration support. As a result, glass ionomers can be used as cavity bases as well as cavity liners.

## Guidelines for basing, lining, and sealing

Clinicians must always consider the limitations of currently available materials. The best possible base for any restoration is sound tooth structure. The following are guidelines for placement of bases, liners, and sealers:

- Do not remove sound tooth structure to provide space for a base. Maintaining sound dentin will enhance restoration support and provide maximum dentinal thickness for pulpal protection.
- Use bases as indicated for buildup materials and block-out materials for cemented indirect restorations. If used for direct amalgam restorations or bonded restorations, minimize the extent of the base. Basing a preparation to “ideal” depth

and internal form is contraindicated.[166] Bases in cavity preparations for amalgam restorations and bonded resin or ceramic restorations lead to decreased bulk of restorative material and increased potential for restoration fracture.

- Use the minimum thickness of liner necessary to achieve the desired result. For liners under amalgam restorations, this should not exceed 0.5 mm.[167]
- While there is research support for the use of adhesive sealers (dentin bonding agents) under amalgam restorations, long-term clinical evidence is not strong.

## Direct and Indirect Pulp Capping

*Pulp capping* is defined as "endodontic treatment designed to maintain the vitality of the endodontium."[168] Several favorable conditions must be present before considering direct or indirect pulp capping[168,169]:

- The tooth must have a vital pulp and no history of spontaneous pain.
- Pain elicited during pulp testing with a hot or cold stimulus should not linger after stimulus removal.
- A periapical radiograph should show no evidence of a periradicular lesion of endodontic origin.
- Bacteria must be excluded from the site by the restoration.

If these conditions can be met, an indirect pulp capping procedure is preferable to a direct pulp capping procedure. An indirect pulp capping procedure allows a protective thickness of dentin to remain adjacent to the pulp. Not only does this provide the advantages previously described for maintaining dentin (as RDT), but because RDT is directly related to odontoblast survival, reparative dentin formation is also enhanced.[3] In addition, avoiding pulp exposure means that there is less chance for infected debris to be introduced into the pulp to cause an inflammatory reaction.[3,170] Avoiding pulp exposure also means that there is no concern for hemorrhage from the pulp, a factor that has been associated with decreased success rates in direct pulp capping.[171–174]

Because of the uncertainty for success with either indirect or direct pulp capping procedures, pulpal health should be monitored for several months in teeth that are to receive indirect restorations or serve as abutments for fixed or removable partial dentures. If the pulpal status of a tooth is uncertain, the clinician should consider endodontic therapy before initiating restorative treatment.

## Direct pulp capping

Considerable research has been done regarding direct pulp capping. However, a caveat is in order because much of this research has used animal models. While providing useful information, pulp capping outcomes in animals often do not reflect the outcomes of pulp capping in humans.[26,175,176] As such, pulp capping findings in animals must be interpreted with caution.

Pulps can be exposed as a result of trauma, caries, or mechanical reasons. The latter exposure is usually of iatrogenic origin. *Direct pulp capping* is an attempt to maintain pulpal vitality by placing a material directly over the exposed pulp. It is hoped that this will allow the pulp to heal normally, regenerate reparative dentin, and prevent the need for more extensive and expensive treatment, such as root canal therapy. Studies have indicated that pulp capping is more likely to be successful if the cause of the pulp exposure is mechanical rather than due to caries.[168,177] A carious exposure will cause bacterial contamination of the pulp resulting in inflammation[131] and a pulp that is less able to respond and heal, whereas mechanical exposure will likely cause less bacterial contamination and resulting inflammation.

After a pulp has been exposed, it is important to control pulpal bleeding before placing a pulp capping agent.[172] Increased bleeding that is difficult to stop can compromise the success of the procedure. This is likely due to one or both of two reasons. First, increased pulpal bleeding may be indicative of a higher level of pulpal inflammation and therefore a reduced capacity for pulpal healing. Second, the blood contamination of dentin adjacent to the exposure site may compromise the seal required to exclude bacterial contamination of the exposed pulp.

Pulpal bleeding is normally controlled with a cotton pellet saturated in a solution applied to the exposure site. A variety of solutions have been used in this situation. Water or saline are the most benign to the pulp.[178] Sodium hypochlorite, in concentrations ranging from 0.12% to 5.25%, is more caustic to the pulp but is extremely effective at controlling bleeding and is very effective at disinfecting the area. It has been used effectively in a number of studies and is likely the most commonly used agent for controlling pulpal bleeding. Chlorhexidine solution is an effectual antibacterial but is not as effective for controlling hemorrhage. Other solutions, such as those used to control gingival bleeding during impression taking, have less evidence to support their use in pulp capping, but short-term data seem to show few adverse effects on the pulp. One exception to this might be ferric sulfate, for which multiple clinical studies have indicated increased postoperative pain when it was used in conjunction with pulp capping.[173,179–184]

It has been suggested that age may have an impact on the success or failure of direct pulp capping, because older pulps have increased fibrosis and a decreased blood supply[185] and thus a decreased ability to mount an effective response to invading microorganisms.[174] However, there is no consensus in the literature regarding age as a factor in the success of direct pulp capping. Some studies have shown no relationship between success and patient age[172,177]; however, other research has found that age did have a bearing on success.[186,187]

The best chance for direct pulp capping to permit formation of a dentin bridge and to maintain pulpal vitality is in the presence of the most ideal conditions. If a large number of bacteria from a caries lesion or exposure to the oral flora have contaminated the pulp, the likelihood of regaining or maintaining a healthy pulp is slight. The adverse consequences of bacterial contamination of the pulp have been well documented.[12–14,23,24] Ideally, direct pulp capping would be attempted only when a small mechanical exposure of an otherwise healthy pulp occurs. A recent systematic review provided evidence that direct pulp capping of carious exposures can be carried out successfully.[188] However, the tooth should exhibit the favorable conditions noted previously. The tooth must be isolated with rubber dam, and adequate hemostasis must be achieved. The exposure should be covered with an appropriate pulp capping agent, and it must be possible to restore the tooth with a well-sealed restoration that will prevent subsequent bacterial contamination (Fig 6-3). The next section will consider the various materials that have been suggested for use as direct pulp capping agents.

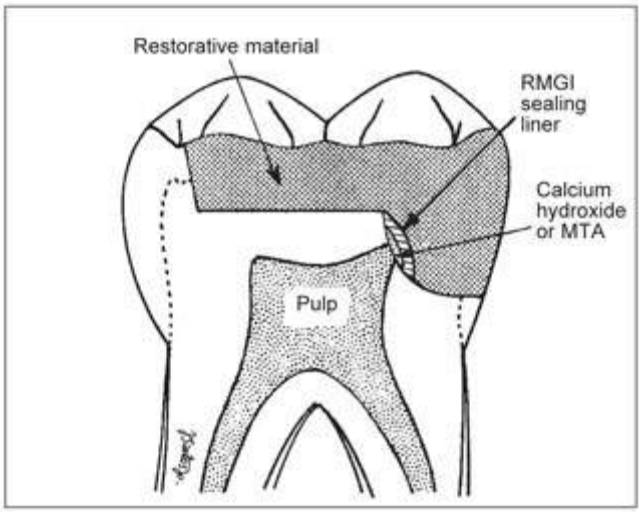



**Fig 6-3** For a direct pulp capping procedure, a calcium hydroxide or mineral trioxide aggregate lining material is placed on the exposed pulpal tissue and a small amount of surrounding dentin. A sealing liner and/or a sealing restoration is then placed to seal out bacteria and their by-products. RMGI—resin-modified glass ionomer; MTA—mineral trioxide aggregate.

## *Direct pulp capping materials*

**Zinc oxide–eugenol.** ZOE has been used in dentistry as a base, liner, cement, and provisional restoration for decades. What is most appealing about ZOE as a potential pulp capping agent is its antibacterial properties. ZOE releases eugenol in concentrations that are quite cytotoxic.[189–192] However, ZOE also exhibits high interfacial leakage.[193] Although it has been suggested that high leakage does not matter because ZOE provides a "bacterial seal" due to the eugenol release, it should be noted that eugenol release declines dramatically with time.[191] This would seem to imply that the effectiveness of ZOE at excluding bacterial penetration would decrease over time.

Clinical research on direct pulp capping with ZOE is sparse. One study found dramatic adverse outcomes for human teeth in which pulps were capped with ZOE. All teeth receiving ZOE pulp caps demonstrated chronic inflammation and a lack of dentin bridging and pulpal healing 12 weeks after pulp capping. However, all control teeth that had been pulp capped with calcium hydroxide showed healing within 4 weeks.[194]

**Glass ionomer/resin-modified glass ionomer.** Glass ionomer and resin-modified glass ionomer demonstrate cytotoxicity when in direct contact with cells, although not to the same degree as ZOE. Resin-modified glass-ionomer formulations tend to show greater cytotoxicity than does the conventional glass-ionomer form. However, both glass ionomer and resin-modified glass ionomer make excellent cavity liners/bases in deep preparations because of their ability to chemically adhere to tooth structure. The resultant seal prevents toxic elements from penetrating into the pulp. Glass ionomer and resin-modified glass ionomer also provide an excellent seal against bacterial penetration, with good biocompatibility when not in direct contact with the pulp.[6,17,192,195,196]

As with ZOE, there is little human clinical research to support direct pulp capping with glass ionomer and resin-modified glass ionomer. One study showed chronic inflammation and lack of dentin bridge formation up to 300 days after direct pulp capping with a resin-modified glass ionomer. The calcium hydroxide–treated control teeth demonstrated significantly better healing.[197]

**Adhesive systems.** In the early 1990s, it was suggested that dentin bonding agents could be used for direct and indirect pulp capping[198] based on the belief that they could provide an effective, permanent seal against bacterial invasion to promote pulpal healing. However, as with ZOE, glass ionomer, and resin-modified glass

ionomer, all components of adhesive systems are cytotoxic to pulpal cells.[199] The cytotoxic effects of the adhesive system components are synergistic with each other, increase with contact time, and are present in single- as well as multicomponent systems. Insufficiently cured adhesives exhibit greater cytotoxicity than adhesives that are well polymerized due to the presence of unpolymerized components.[199,200]

The interest in using adhesives for pulp capping was driven, at least in part, by the fact that some studies on nonprimates found that mechanical pulp exposures capped with adhesives generally resulted in pulpal healing.[201–203] These results were not unanimous, as some nonprimate studies did find inferior healing following pulp capping with adhesives compared with calcium hydroxide.[204,205] A number of studies on primates with noncontaminated mechanical pulp exposures capped with adhesive systems generally resulted in healing comparable with that of calcium hydroxide.[171,206–208] However, the outcome changes when the results are examined from studies of bacteria-contaminated mechanical pulp exposures in primates. This experimental regimen was chosen to more closely resemble the situation that might be encountered if a pulp exposure occurred due to caries or without a rubber dam in place. These contaminated exposures capped with adhesives resulted in poor pulpal healing compared with that of calcium hydroxide.[173,207]

When the results of human pulp capping studies are reviewed, the conclusions become very different than what would have been deduced from animal studies. Table 6-2 summarizes several human studies comparing pulp capping with calcium hydroxide versus adhesives. In each study cited, calcium hydroxide provided significantly improved pulpal repair compared with adhesive systems, regardless of whether it was an etch-and-rinse or self-etching system. There are several possible explanations for these poor outcomes in human studies. First are the direct cytotoxic effects produced by adhesives on pulpal cells.[199] Second is the difficulty in obtaining an adequate seal to protect against bacterial contamination. This poor seal may be due to one or more reasons. Etchant and primer components of adhesives are vasodilators, which can increase bleeding that contaminates adjacent dentin and degrades adhesion.[180,214,215] The increased moisture at the pulp capping site reduces polymerization of the adhesive. This has the dual detrimental effects of decreasing adhesion and increasing the availability of the unpolymerized and more toxic components of the adhesive.[216] Finally, resin components reduce the pulp's immune response, making it less likely that the pulp will be able to defend itself against bacterial contamination.[217] These findings were confirmed in a review of pulp capping with adhesives, in which De Souza Costa and others[175] concluded the

following: *(1)* adhesives result in inferior pulp healing; *(2)* adhesives result in chronic inflammation, even in the absence of bacteria; and *(3)* inflammation is a poor environment for pulpal healing.

| Table 6-2 | Human study outcomes comparing direct pulp capping with calcium hydroxide versus adhesive systems* | | | |
|---|---|---|---|---|
| **Study†** | **No. of teeth** | **Restoration** | **Time (mo)** | **Results‡** |
| Accorinte et al, 2006[209] | 40 | Etch-and-rinse/composite | 2 | Calcium hydroxide |
| De Souza Costa et al, 2001[26] | 33 | Self-etch/composite | 10 | Calcium hydroxide |
| Accorinte et al, 2008[210] | 34 | Self-etch/composite | 3 | Calcium hydroxide |
| Sübay and Demirci, 2005[211] | 16 | Etch-and-rinse/composite | 1 | Calcium hydroxide |
| Accorinte et al, 2005[212] | 25 | Etch-and-rinse/composite | 2 | Calcium hydroxide |
| Fernandes et al, 2008[213] | 46 | Etch-and-rinse/composite | 1 | Calcium hydroxide |
| Hörsted-Bindslev et al, 2003[214] | 34 | Etch-and-rinse/composite | 2 | Calcium hydroxide |

*Reprinted from Hilton[169] with permission.
†All studies used mechanical exposures and performed histologic analysis.
‡Results column denotes which pulp cap material performed better.

Another issue in placing bonding resins directly on pulpal tissue is heat generation from a quartz-tungsten-halogen (QTH) or a light-emitting diode (LED) curing light. An intrapulpal temperature increase of more than 20°F (11.2°C) has been shown to cause irreversible damage in vivo.[218] One study investigated the temperature rise in a bonding resin during polymerization with a QTH curing light.[219] An increase of 18.2°C was found with a 10-second cure, and a 25.2°C increase was detected with a 20-second cure. Because most bonding resins require a 10-second cure, and some a 20-second cure, there is potential for the pulp to be exposed to dangerous heat levels. It was originally hoped that the LED curing lights would be shown to produce less heat than QTH lights. This was the case when first introduced, with LED lights demonstrating significantly less heat production than plasma arc curing

(PAC) lights and high-intensity QTH (HQTH) lights.[220] However, current LED lights demonstrate considerably higher light intensity output and are capable of generating much greater heat. This can be a biologic hazard to the pulp when little or no remaining dentin is present.[221]

**Calcium hydroxide.** Calcium hydroxide was introduced to the dental profession in 1921 and has been considered the gold standard of direct pulp capping materials for several decades for a number of reasons.[186] Calcium hydroxide has excellent antibacterial properties.[222] One study found a 100% reduction in microorganisms associated with pulpal infections after 1 hour of contact with calcium hydroxide.[223] Most important, calcium hydroxide has a long-term track record of clinical success as a direct pulp capping agent in periods of up to 10 years,[172,209,224] although reduced success rates have been found in studies in which dental students were the operators.[168,177,214,225]

Calcium hydroxide has some disadvantages as well. The self-curing formulations are highly soluble and are subject to dissolution over time,[226] although it has been noted that by the time the calcium hydroxide is lost due to dissolution, dentin bridging has occurred.[209] Calcium hydroxide has no inherent adhesive qualities and provides a poor seal.[227] Another criticism noted of calcium hydroxide is the appearance of so-called tunnel defects in reparative dentin formed beneath calcium hydroxide pulp caps.[228,229] A *tunnel defect* has been described as a patency from the site of the exposure through the reparative dentin to the pulp, sometimes with fibroblasts and capillaries present within the defect.[229] Other researchers have found that the quality of reparative dentin improves as the bridge gets thicker[230] and that the tunnel defects are not often patent with the pulp.[204] Of studies of direct pulp capping with calcium hydroxide, more did not report tunnel defects than did report them.[180,209,210,212,214]

Calcium hydroxide is believed to effect pulpal repair by one or more of several mechanisms of action. Calcium hydroxide possesses antibacterial properties, and this can minimize or eliminate bacterial in the capped pulp.[222] Traditionally, it has been believed that calcium hydroxide's high pH causes irritation of the pulpal tissue, which stimulates repair via some unknown mechanism.[132] In recent years, this "unknown mechanism" has been explained as the release of bioactive molecules. It is known that a variety of proteins are incorporated into the dentinal matrix during dentinogenesis. Of particular importance to the topic of pulp capping, at least two of these proteins, bone morphogenic protein (BMP) and transforming growth factor-β1

(TGF-β1), have demonstrated the ability to stimulate pulpal repair.[231–233] Furthermore, calcium hydroxide is known to solubilize these proteins from dentin, lending credence to the release of these bioactive molecules as a significant mediator in pulpal repair following pulp capping.[132,233]

**Mineral trioxide aggregate.** Mineral trioxide aggregate (MTA) has generated considerable interest as a direct pulp capping agent in recent years. Unset MTA is primarily calcium oxide in the form of tricalcium silicate, dicalcium silicate, and tricalcium aluminate. Bismuth oxide is added for radiopacity.[234,235] MTA is considered a silicate cement rather than an oxide mixture, and so its biocompatibility is due to its reaction products.[236] Interestingly, the primary reaction product of MTA with water is calcium hydroxide.[234,236–238]

As a result, many of the advantages and potential mechanisms of action for MTA are similar to those for calcium hydroxide, including its antibacterial and biocompatibility properties, high pH, radiopacity, and its ability to aid in the release of bioactive dentin matrix proteins.[237–243] There are some differences between MTA and calcium hydroxide. First, MTA is available in two colors, white and gray. The gray version is created by the addition of iron.[244] Another significant difference is the fact that MTA does provide some seal to tooth structure.[245]

There are several disadvantages to the use of MTA as well. It has shown high solubility, demonstrating 24% loss after 78 days' storage in water.[237,238] The presence of iron in the gray MTA formulation may darken the tooth.[242] A significant downside to MTA is the prolonged setting time of approximately 2 hours, 45 minutes.[240,241] This requires that pulp capping with MTA either be done in a two-step procedure, placing a provisional restoration to allow the MTA to set before placing the restoration, or using a quick-setting liner to protect the MTA during definitive restoration placement. The handling characteristics of the powder-liquid MTA are very different from the typical paste-paste formulations of calcium hydroxide that most practitioners find easy to handle. When compared with these paste-paste formulations of calcium hydroxide, MTA is very expensive. One gram of MTA powder costs approximately the same as 24 grams of calcium hydroxide base/catalyst paste, making MTA much less cost-effective per use.

A review of direct pulp capping studies in animals comparing MTA with calcium hydroxide generally reveals better pulpal healing with MTA.[246–250] As with pulp capping studies comparing adhesives with calcium hydroxide, the results are different when comparing MTA with calcium hydroxide in human studies. Most

human studies show similar outcomes of MTA and calcium hydroxide.[242,251–255] However, four studies do demonstrate superior performance of MTA.[251,256–258] These studies must be interpreted cautiously, because two of them are very short term (2 and 3 months),[251,256] and both share an interesting study characteristic: The pulp capped teeth were restored with a temporary ZOE material, while a definitive restoration was used in the other studies. As discussed in the section on ZOE, these materials leak significantly and lose their antibacterial eugenol-release rapidly. Thus, these results may highlight the ability of MTA to provide a seal over the pulp exposure that calcium hydroxide does not. The third study also suffers from some shortcomings, including the facts that the treatment was not randomly assigned and the study had too few subjects to adequately determine the outcome with appropriate power.[257] The fourth study was a practice-based trial of pulp caps for mostly carious pulp exposures with 358 participants followed for up to 2 years, which also showed significantly better outcomes with MTA compared with calcium hydroxide. A recent systematic review[188] showed significantly better outcomes for MTA with an indirect comparison between calcium hydroxide and MTA on the weighted pooled success rate. Additional human studies using MTA as the sole pulp capping agent with no control group have shown good success in periods ranging from 6 months to 4 years.[183,259,260]

On the basis of the literature to date, it would appear that MTA's success is likely due to the fact that it serves as a reservoir for calcium hydroxide and/or its capacity to provide a seal at the site of the pulp exposure. Even though MTA seals better than calcium hydroxide, it should be kept in mind that glass ionomer or resin-modified glass ionomer will be needed as a liner over either pulp capping material. In the case of calcium hydroxide, the glass-ionomer/resin-modified glass-ionomer liner is needed to provide a protective antibacterial seal that the calcium hydroxide alone cannot provide. In the case of MTA, the glass-ionomer/resin-modified glass-ionomer liner is needed to protect the MTA during restoration placement due to the prolonged setting time. Without this glass-ionomer/ resin-modified glass-ionomer protective sealer, it would be necessary to place a provisional restoration for a period of time until the MTA is set, requiring a second appointment for definitive restoration placement.

**Success in direct pulp capping.** With its long-term track record of clinical success, calcium hydroxide has long been considered the gold standard for direct pulp capping. However, the sum total of the latest research comparing calcium hydroxide and MTA shows that MTA demonstrates superior outcomes for direct pulp capping.

There are several keys to direct pulp capping success: restricting pulp capping to asymptomatic or mildly symptomatic teeth consistent with reversible pulpitis, controlling bleeding, providing a bacterial seal at the exposure site, and providing a well-sealed restoration following the pulp capping procedure.

## Indirect pulp capping

Removal of carious tooth structure has been a basic step in dentistry. When the caries lesion is deep, every restorative dentist is faced with the decision as to the best way to proceed: Is it better to remove all carious dentin regardless of pulpal consequences or to stop short to avoid pulp exposure? When practitioners in a dental practice-based research network were given a hypothetical scenario that involved this question, only 17% responded that they would stop, leave the remaining carious dentin in place, and restore the tooth.[261] This procedure, in which carious dentin is allowed to remain adjacent to a vital pulp—rather than risk pulp exposure—and is covered with a cavity sealer or liner prior to restoration is termed an *indirect pulp capping procedure*, or *indirect pulp treatment*[262] (Fig 6-4). Pierre Fauchard first suggested indirect pulp capping 200 years ago.[263]

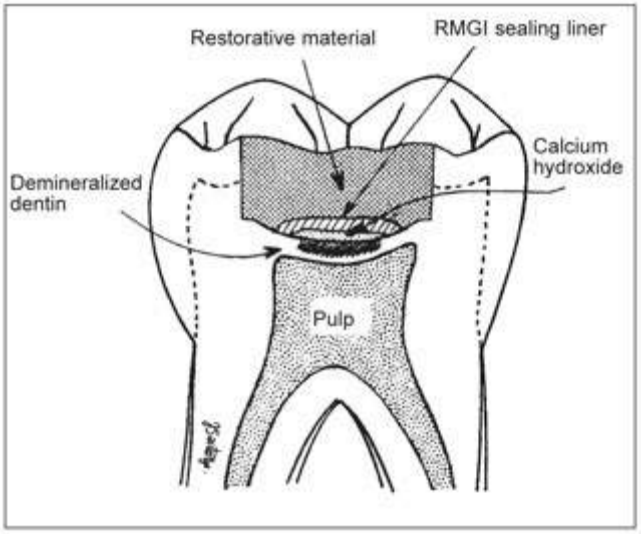



**Fig 6-4** In an indirect pulp capping procedure, all carious, demineralized dentin is removed in the periphery of the preparation, but a small amount of demineralized dentin is left immediately over the area of the pulp. A calcium hydroxide lining material is placed to cover the remaining demineralized dentin. A sealing liner and/or a sealing restoration is then placed to seal out bacteria and their byproducts. RMGI—resin-modified glass ionomer.

The evidence regarding indirect pulp capping stands in contrast to the response of the practitioners, however. Several studies show that restored teeth with partial caries removal have equal success compared with restored teeth with complete

caries removal.[264–266] A number of studies have evaluated the fate of caries lesions in which partial caries removal was accomplished. Typically, an initial clinical and microbiologic assessment of the caries lesion is carried out, partial caries removal is accomplished, and a sealer or liner and restoration is placed for a period of 4 to 12 months before the tooth is reentered and reassessed. Invariably, these studies find that *(1)* the lesion color has changed from light brown to dark brown, *(2)* the tissue consistency has changed from soft and wet to hard and dry, *(3)* *Streptococcus mutans* and *Lactobacillus* have been significantly reduced to a limited number or even zero viable organisms, and *(4)* the radiographs show either no change or even a decrease in the radiolucent zone.

It is known that inflammation is present when active caries is adjacent to the pulp, but there is little to no inflammation if caries is arrested.[267] The type of liner is less important to success than the placement of a well-sealed restoration.[266,268–275] In addition, partial removal of carious dentin significantly reduces the chance of pulp exposure during caries excavation.[276,277] This is very important, because pulp exposures result in more pulp injury, inflammation, and poorer prognosis for pulp vitality than when pulp exposure is avoided.[278,279]

Two thorough systematic reviews confirmed the scientific validity of indirect pulp capping and concluded the following: *(1)* partial removal of carious dentin reduced the risk of pulp exposure by 98% compared with complete caries excavation in teeth with deep caries lesions; *(2)* there is no evidence that partial removal of carious dentin is detrimental in terms of signs, symptoms, pulpitis occurrence, or restoration longevity; and *(3)* there is substantial evidence that complete removal of carious dentin is not needed for success provided the restoration is well sealed.[280,281]

An indirect pulp capping procedure (Box 6-1) should be considered when there is a radiographically or clinically evident deep caries lesion encroaching on the pulp and the tooth has no history of spontaneous pain and responds normally to vitality tests. Pulp exposure must be avoided; if it occurs, it should be regarded as an iatrogenic event. A direct pulp capping procedure should be necessary only if the operator inadvertently exposes the pulp in attempting an indirect pulp capping procedure. With a deep caries lesion, the indirect pulp capping procedure is always preferred to a direct pulp capping procedure.

| **Box 6-1** | **Protocol for indirect pulp capping procedures** |
|---|---|
| **Diagnosis** | |

The preoperative status of the pulp and periradicular tissues should be evaluated carefully. The tooth is a candidate for indirect pulp capping only if the following conditions exist:

- There is no history of spontaneous pulpal pain.
- Pulpal vitality has been confirmed with thermal or electric pulp testing.
- Pain occurring during application of a hot or cold stimulus does not linger after the tooth returns to mouth temperature.
- A periapical radiograph shows no evidence of a periradicular lesion of endodontic origin

**Treatment**

1. Isolation: After administering anesthetic, isolate the tooth with a rubber dam.
2. Preparation: Prepare the tooth for the definitive restoration, leaving demineralized dentin only in the area immediately adjacent to the pulp. All peripheral carious tooth structure, particularly at the cavosurface margins, must be removed. Use a caries-disclosing dye if necessary to ensure complete carious dentin removal (other than that immediately adjacent to the pulp). After this is accomplished, use a spoon excavator or a large round bur in a low-speed handpiece at very low speed.

   Use very gentle, featherweight strokes over the area of the demineralized dentin to remove only the wet (soft, amorphous) carious dentin. Leave the dry,fibrous, place a layer of resin-modified glass ionomer that covers the calcium hydroxide and extends onto sound dentin on the periphery to provide a seal.
4. Restoration:
   a. Direct restorations: For direct restorations (bonded amalgam, resin composite, glass ionomer), place the definitive restoration. If time does not allow for placement of a definitive restoration at the first appointment, a glass-ionomer or reinforced ZOE provisional restoration should be placed and another appointment scheduled for the definitive restoration as soon as possible. The indirect pulp capping liner should not be disturbed during the subsequent restoration process.
   b. Indirect restorations: For indirect restorations (cast metal restorations, ceramic onlays, or crowns), place a definitive buildup (bonded amalgam, resin composite, glass ionomer) at the appointment in which the indirect pulp capping procedure was performed if time allows. Delay placement of the definitive restoration for 4 to 8 months. Before proceeding with the definitive restoration, ensure normal pulp vitality.

**Precautions**

- Use care in removing carious dentin near the pulp to prevent accidental pulp exposure.
- If a provisional restoration has previously been placed over an indirect pulp capping liner and the tooth is reentered for a restorative procedure, do not remove the indirect pulp capping material.
- Prior to excavation, use tactile exploration to confirm that dye-stained dentin lacks hardness.

| | |
|---|---|
| demineralized dentin that gives some moderate resis<br>tance to gentle scraping with a spoon excavator.<br>3. Lining: Place a calcium hydroxide liner over the remain<br>ing demineralized dentin and allow it to set. Then | |

For indirect pulp capping (Figs 6-5 and 6-6), after the initial entry into the carious dentin (see Fig 6-5c), a spoon excavator or large round bur, rotating very slowly in a low-speed handpiece, should be used to excavate the caries-softened dentin. Demineralized dentin not near the pulp should be completely removed, leaving hard, sound dentin. As the excavation of carious dentin nears the pulp, caution must be exercised to avoid pulp exposure. A spoon excavator may aid in tactile detection of softened dentin. The wet (soft, amorphous) carious dentin should be removed; as the pulp is approached, the dry, fibrous, demineralized dentin that offers some moderate resistance to gentle scraping with a spoon excavator should be allowed to remain.

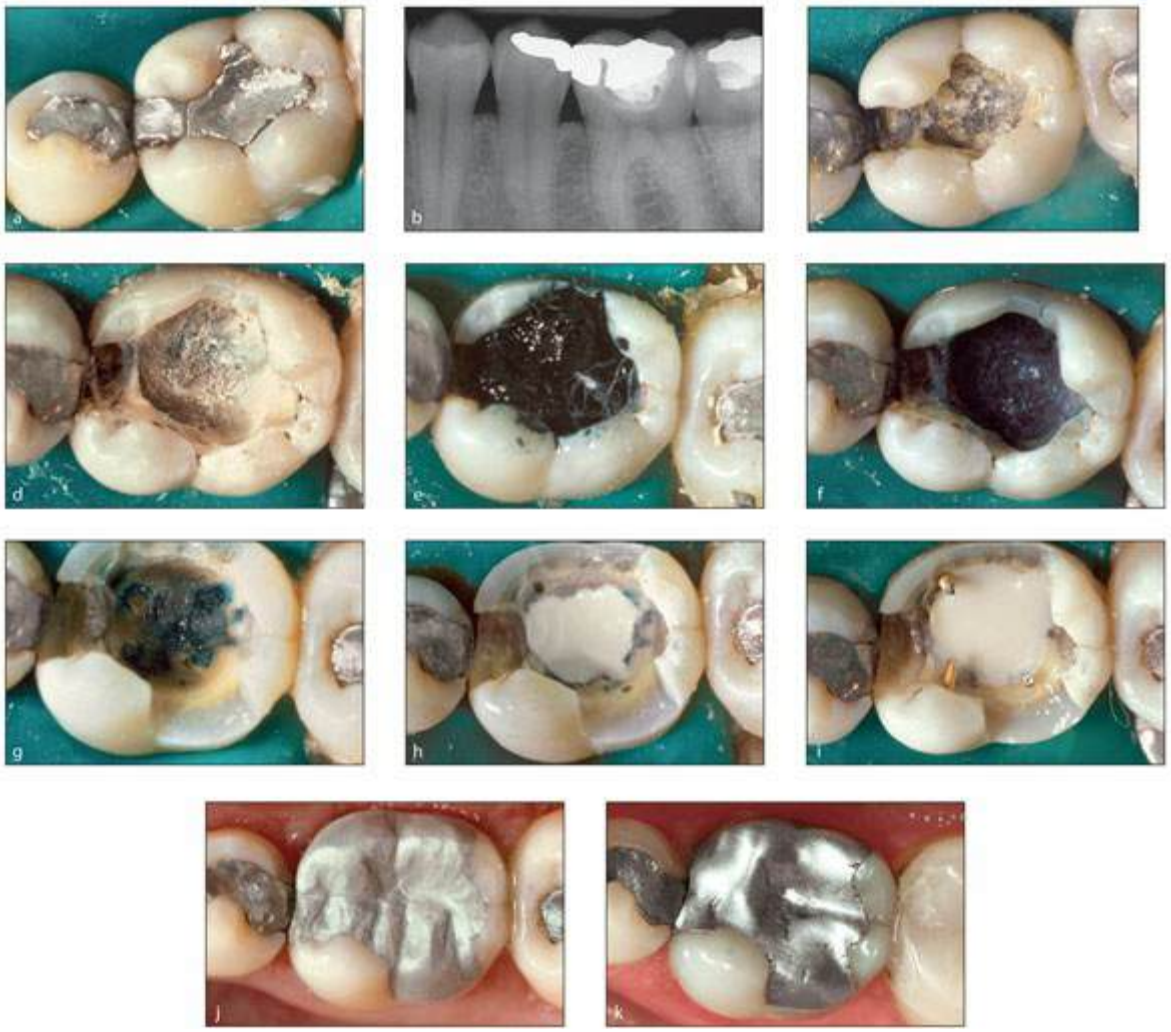


**Fig 6-5** Indirect pulp capping procedure, mandibular left first molar. *(a)* Mandibular first molar with deep recurrent carious involvement. Preoperative evaluation indicated vital pulpal tissue and no history of pain. *(b)* Preoperative radiograph shows demineralized dentin around the base, under the amalgam, with carious demineralization advancing toward the pulp. *(c)* The old restoration and base are removed, revealing soft, carious dentin. *(d)* The preparation is widened to enable removal of a large amount of carious dentin and some undermined enamel. *(e)* A cotton pellet, saturated with a blue caries-disclosing dye solution (Cari-D-Tect, Gresco), is placed so that the dye coats all areas of the preparation. The cotton pellet is left in place for 10 seconds and removed. The area is then washed with air-water spray. *(f)* Note the remaining large amount of demineralized dentin, as revealed by the dye. The stained dentin is checked for softness, and softened dentin is removed with a large round bur rotating very slowly. The dye is reapplied and rinsed, and additional softened dentin is removed. *(g)* Most of the demineralized dentin has been removed. Only a small amount of dye-stained dentin is left; this dentin is believed to immediately overlie the pulp. *(h)* A calcium hydroxide liner (Dycal, Caulk/Dentsply) is placed over the remaining demineralized dentin. *(i)* A glass-ionomer liner (Vitrebond, 3M) is placed over the calcium hydroxide liner. The glass-ionomer liner provides some seal and improved strength for amalgam condensation. *(j)* The completed amalgam restoration immediately after placement. *(k)* The restoration after 7 years of service.

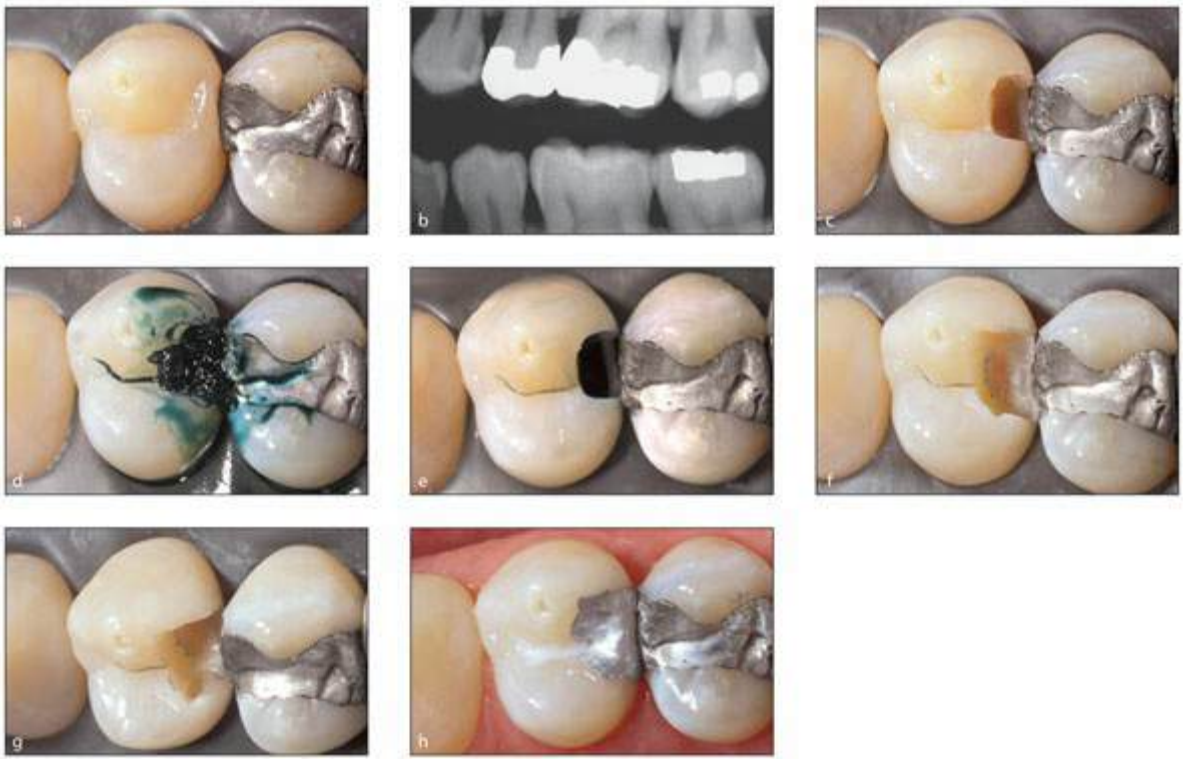


**Fig 6-6** Indirect pulp capping procedure, maxillary left first premolar. *(a)* Maxillary left first premolar with extensive proximal caries on the distal surface. *(b)* Bitewing radiograph reveals the extent of the caries lesion. *(c)* Initial entry reveals soft, carious dentin. *(d)* Caries-disclosing dye is applied. *(e)* Dye-stained carious dentin. *(f)* After several applications of caries-disclosing dye and removal of caries-softened dentin, only a small amount of dye-stained dentin remains adjacent to the pulp chamber. *(g)* Calcium hydroxide and glass-ionomer liners are applied over the remaining demineralized dentin. *(h)* The completed amalgam restoration. The deep and stained fissure was addressed using a pit and fissure sealant.

Caries-disclosing dyes may be used to assist in excavation of carious dentin (see Figs 6-5e to 6-5g and 6-6d to 6-6g). Studies have demonstrated the benefit of these dyes to aid in identification and removal of demineralized dentin[282] and to greatly reduce remaining viable bacteria.[283] It must be recognized that the dyes stain not only demineralized dentin but also anything porous, such as debris that may have been left in the cavity preparation. In addition, noncarious deep dentin will absorb the dye[284–286] because of the increased number and size of the dentinal tubules in deep dentin; if this dye-stained sound dentin is removed, pulp exposure will result. There has been concern that, when using a dentin bonding system, the previous use of caries-disclosing dyes can reduce the strength of the bond to dentin and increase microleakage at the interface of the bonded restoration with the wall of the cavity.[287] However, the preponderance of evidence indicates that neither of these should be a concern.[288–290]

In the indirect procedure, all carious dentin is removed except for the last portion

of firm, leathery carious dentin immediately overlying the pulp. At this point, a calcium hydroxide liner is placed over the demineralized area of dentin (see Fig 6-5h). Placement of calcium hydroxide over this layer of leathery dentin has been shown to virtually eliminate all remaining bacteria and to render the residual carious dentin operationally sterile.[291,292] Then a layer of resin-modified glass ionomer is placed, covering the calcium hydroxide and extending onto sound dentin on the periphery to provide a seal (see Fig 6-5i). Resin-modified glass ionomer alone (without calcium hydroxide as an initial layer) is also effective at providing favorable clinical and microbiologic changes when used as a liner on remaining caries.[275] If any vital bacteria remain, a well-sealed restoration should isolate them from life-sustaining substrate and prevent further acid production, thereby arresting the caries process.[264,293] These facts argue against a two-step procedure in which the tooth is reentered for the purpose of excavating the remaining acid-affected dentin to confirm reparative dentin formation. Bacterial levels in caries lesions are reduced most significantly after the initial caries excavation and liner placement, with little additional benefit from the subsequent excavation.[71] In addition, a second caries removal procedure risks creating a pulp exposure and causing further traumatic insult to the pulp.[294] A glass-ionomer liner should be placed over the calcium hydroxide liner to improve strength during amalgam condensation and to enhance the seal (see Figs 6-5i and 6-6g). The definitive restoration should be placed to minimize microleakage at the interface of the restoration with the cavity-preparation walls (see Figs 6-5j and 6-6h).

## The future of direct pulp capping materials

A number of materials are being investigated for future use as direct pulp capping agents. Hydroxyapatite elicited a better pulpal response than calcium hydroxide in one animal study because the hydroxyapatite acted as a scaffold for dentin formation.[295] It seems that the most promising research area in pulp capping likely relates to placing materials that either contain bioactive molecules or are capable of releasing them from the dentin matrix. Growth factors such as BMP and TGF-β1 offer intriguing potential as initiators and/or mediators in pulpal repair.[233] In an animal study, BMP and bone sialoprotein (BSP) were more effective for inducing reparative dentin than was calcium hydroxide.[296] Another animal study showed that TGF-β1 was able to stimulate pulpal stem cells to differentiate into odontoblast-like cells that produced reparative dentin.[232]

## Antibacterial Efficacy of Restorative Materials

One of the keys to any successful restoration, but particularly for a tooth that has undergone a pulp capping procedure, is the ability to exclude bacteria and their by-products from entry into the pulp.[13,168] Of particular concern are the cariogenic bacteria, which tend to invade the interface between the restoration and the cavity preparation. Ultimately, these bacteria may lead to recurrent caries, and, if they reach the pulp in sufficient quantities, they will cause inflammatory and ultimately pathologic responses.[13] In particular, *S mutans* is often used in in vitro studies evaluating the efficacy of restorative materials against bacteria, because this organism is associated with recurrent caries[297] (see chapter 5).

The discussion so far has focused on the materials placed immediately adjacent to the site of near or actual pulp exposure. However, the seal provided by the restorative material will also affect the ultimate success of the procedure. Restorative materials can effectively prevent bacterial contamination by one of two means: *(1)* providing an impermeable seal with the cavity walls to physically exclude bacteria and the toxins they produce or *(2)* possessing antibacterial properties to destroy bacteria entering the restoration-tooth interface. No material yet provides an impermeable seal that can ensure the physical exclusion of bacteria.

## Amalgam

Although typically not considered a material possessing antibacterial properties, dental amalgam has demonstrated varying levels of antibacterial activity.[297–301] This activity has been attributed to a variety of elements released from amalgam, including copper, mercury, zinc, silver, and chloride compounds. A number of studies have shown that amalgam is effective against cariogenic bacteria, including *S mutans, Actinomyces viscosus,* and *Lactobacillus* species.[297–301]

*S mutans* thrive and produce lactic acid in an acidic environment.[302] The tooth-amalgam interface has a decreased pH,303 which results in demineralization of tooth structure.[304] In vivo studies have shown that metallic solutions of copper, silver, and zinc are all capable of reducing acid production in plaque.[305] In one case, ions of these metals reduced acid production more than did fluoride in a comparable concentration.[306] All of these ions are released by amalgam and would therefore be present at the tooth-restoration interface.[298–301]

In addition to its antibacterial properties, amalgam is the only restorative material in which the marginal seal improves with time. This is due to the acidic environment

and low oxygen concentration that exists in the amalgam-to–cavity wall gap, which promotes corrosion. In conventional amalgam, the gamma-2 phase forms $SnO_2$, $SnCl_2$, and $Sn(OH)_2Cl$, which slowly fill the interfacial gap. In high-copper alloys, in which there is no gamma-2 phase (tin-mercury phase), the eta phase (copper-tin phase) corrodes to form $CuO_2$ and $CuCl_2$, but this occurs much more slowly. Corrosion in high-copper alloys may take twice as long as in conventional alloys to produce the same level of seal.[307]

## Glass ionomer

Glass ionomer has the ability to decrease bacterial penetration,[145] possibly through its fluoride release, initial low pH,[147] physical exclusion of bacteria,[145] or release of a metal cation.[148,149] Whatever the mechanism, glass-ionomer restorative and liner/base materials inhibit cariogenic bacteria[151] and demineralization at tooth-restoration interfaces.[308,309] In vivo plaque studies assessing the level of cariogenic bacteria invariably show significantly lower levels of these organisms adjacent to glass ionomer compared to either resin composite or amalgam.[310–312]

## Resin composite

In contrast to amalgam and glass ionomer, resin composite is most dependent on the formation of an impermeable seal to exclude bacterial penetration. This is because, as shown in in vitro bacterial-inhibition studies, there is little, if any, inhibitory effect demonstrated by resin composite against cariogenic bacteria,[149] and, therefore, there is little resistance to secondary caries activity.[309] This is true even if a resin composite contains fluoride.[309] In fact, research has indicated that certain monomers released from resin composite actually stimulate cariogenic bacterial growth.[313] In vivo plaque studies have demonstrated that levels of cariogenic bacteria in the plaque present on surfaces of resin composite restorations are significantly higher than on either amalgam or glass ionomer.[311–315] While certain adhesive systems are similar to resin composite in that they demonstrate no bacterial inhibition,[316] some glutaraldehyde-containing bonding systems have shown an inhibitory effect on cariogenic organisms.[316,317] As previously stated, the quality and durability of adhesive bonding to dentin is questionable, and an impermeable seal may not be achieved. Because resin composite does not have the ability to inhibit cariogenic bacteria, placement of a resin composite restoration with a margin in

dentin in a tooth that has been treated with pulp capping may decrease the chances of successful treatment.

## Summary

- Most operative procedures are traumatic to the pulp, and the effects are at least somewhat cumulative. Excessive heat and dehydration should be avoided. Questionable teeth should receive pulp vitality testing before undergoing clinical procedures.
- Avoid exposing the pulp. The chances for tooth survival are excellent if the tooth is asymptomatic and well sealed, even if residual carious dentin remains.
- Control hemorrhage with water, saline, or sodium hypochlorite. Water and saline are the most benign to the pulp; sodium hypochlorite is more effective for controlling hemorrhage and disinfecting.
- Because no dentin remains between the capping material and the pulp for direct pulp capping, the problem of exposure of pulpal tissues and surrounding vital dentin to caustic or toxic materials is significant. The effects of thermal and chemical insults are magnified with an exposed vital pulp. While calcium hydroxide has been the gold standard for pulp capping because of its long track record of clinical success, MTA has demonstrated superior performance to calcium hydroxide in recent clinical studies. Either calcium hydroxide or MTA should be used with very meticulous clinical technique and excellent isolation. Indirect pulp capping is preferred to direct pulp capping; with proper diagnosis and good clinical procedure, direct pulp capping should rarely be required.
- Calcium hydroxide or glass ionomer are the materials of choice for indirect pulp capping. If the restoration-to–cavity wall interface is well sealed, calcium hydroxide eliminates or greatly reduces the number of viable bacteria in the remaining demineralized dentin. Dentin bonding resins adhere poorly to carious dentin, provide an inadequate seal, and impart little to no antimicrobial activity.
- Drawbacks attributed to the use of calcium hydroxide as a pulp capping agent include dissolution with acid etching, degradation under leaking restorations, and interfacial failure during amalgam condensation. While the most significant drawback of calcium hydroxide is its inability to provide a permanent seal against bacterial invasion, the integrity and durability of the bond achieved with dentin adhesives is questionable as well. ZOE, glass ionomer, resin-modified glass ionomer, and adhesives are poor direct pulp capping agents and should be

avoided for this application.

- When calcium hydroxide or MTA is used as a pulp capping material, it should be limited to as small an area as possible, and some method of protecting it should be considered during subsequent restorative procedures. Placement of a glass-ionomer lining material over the calcium hydroxide provides a combination of clinically proven materials associated with clinical success in pulp capping. Calcium hydroxide provides antibacterial properties; glass ionomer, especially resin-modified glass ionomer, provides resistance to acids, condensation pressures, and dissolution, as well as adhesion to the tooth structure and a source of fluoride in the event of leakage.
- Provide a well-sealed restoration immediately after pulp capping. This will provide protection against ongoing leakage and bacterial contamination that can compromise success.

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